

Methodological document

Pathway and methodology of the DCEN Report v3

Simulation 3 - DCEN + EducationLibre + European ENR Coalition

- How the model was built, step by step
- Intellectual decisions and their justifications
 - Corrections made and why
- Sources consulted and how they were integrated
 - Design principles retained

This document traces the thinking process, not the result.
Its purpose is to make the DCEN model understandable, criticisable and improvable.

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1. Starting point - Theoretical foundations

The DCEN model did not emerge from a blank page. It was built from a simple question posed at the start of the conversation: what would the combination of Piketty, Saez and Zucman with the SSE produce? This question was the guiding thread of all subsequent construction.

1.1 The three economists as diagnosticians

The first methodological decision was to distinguish diagnosis from prescription. Piketty, Saez and Zucman are diagnosticians: they measure, map and document inequalities. They do not prescribe a complete alternative economic model. The DCEN is the prescription - distinct from the diagnosis but grounded in it.

Economist	Contribution to diagnosis	Identified limit
Thomas Piketty	Theoretical framework $r > g$: capital self-amplifies without intervention	Remains within a redistribution logic, not structural transformation
Emmanuel Saez	Empirical evidence: inequalities are a documentable political choice	US focus - European extrapolation must be handled with caution
Gabriel Zucman	Mapping of offshore evasion: 10% of world GDP invisible to tax authorities	Blockchain as solution is not in his work - it is an extension

1.2 The SSE as an alternative productive structure

The second decision was not to confuse redistribution with transformation. Piketty/Saez/Zucman deal with redistribution. The SSE deals with who produces and how. Both are necessary and complementary, but they answer different questions. This distinction structured the entire architecture of the model.

2. Iterative construction of the DCEN model

The model was not conceived as a whole. It was built through successive additions, with each new building block tested for coherence with the preceding ones. Here are the steps and the questions that guided each addition.

1

Piketty + Saez + Zucman + SSE

Diagnostic foundation + productive structure

Initial question: what happens if we combine the inequality diagnosis with a democratic productive structure? The answer: it works, but the fiscal capture and investment mechanism is missing. Hence the next step.

2

Adding the citizens sovereign fund

The common pot that invests

Question: who invests the money recovered through taxation? A classical state risks recentralisation. A sovereign fund with democratic governance resolves this tension. Reference models: Norway, Alaska, Singapore.

3

Adding reversed cantonal taxation

The fundamental Swiss innovation

Question: how to make progressive taxation politically acceptable in a federal context? The answer: reverse the chain. Cantons collect and pass on, rather than the state collecting and distributing. The 40/35/25 key as the balance of subsidiarity and solidarity.

4

Adding the global blockchain

Making wealth visible

Question: how to make global taxation operational without tax havens? The blockchain as a universal wealth register - not Zucman's idea but a logical extension of his work on traceability. Important distinction: fiscal blockchain (transparency) vs cryptocurrency (speculation).

5

Adding 4-day week + inclusion

Reconfiguring human time

Question: where do citizens find the time to govern democratically? The 4-day week as the enabling condition for SSE governance and P2P education. Without free time, participatory democracy remains a slogan.

6

Adding P2P education

The epistemic cement

Question: how to ensure citizens understand what they are governing? P2P education as the epistemic infrastructure of the model. Without it, the SSE remains governed by external experts despite good intentions.

3. Simulation 1 - Green energy transition

The first simulation was designed to test the internal coherence of the model against a concrete challenge: the transition to renewable energy. The goal was not to produce exact figures but to verify that the DCEN building blocks were logically articulated with a real energy constraint.

3.1 Methodological choices of the simulation

Three temporal phases rather than a single figure

Modelling in phases (0-5 years, 5-15 years, 15-30 years) shows the dynamics of the transition and identifies tipping points. A single figure masks the non-linearity of the process. The phases correspond to political (electoral mandates), economic (investment cycles) and systemic (cultural transformation) horizons.

Interactive parameters rather than a fixed scenario

The interactive simulator (with sliders) was chosen to show the sensitivity of the model to assumptions. There is no single optimal scenario - there is a space of valid configurations depending on political priorities. This makes the simulation honest: it does not claim to predict.

Identifying the critical tension: fossil/ENR sequencing

The key finding of simulation 1: exiting fossil fuels BEFORE renewables reach 70% of the energy mix creates a risk of shortage. This sequencing is not trivial and illustrates why the transition must be coordinated rather than simply taxing carbon (the 2018 French mistake).

4. Simulation 2 - European ENR coalition

Simulation 2 emerged from a question about the supranational dimension: if the world's heads of state and parliaments agreed on a global blockchain with taxation making super-wealth morally indefensible, what would happen? The European coalition was identified as the realistic transition scale between the national model and a global agreement.

4.1 The qualitative shift of the coalition

The key methodological decision of simulation 2 was to show that the coalition is not a simple addition of countries but a regime change. Three emergent effects specific to the coalition - impossible at national scale - structured the simulation:

- Systemic energy independence (end of Russian gas blackmail)
- Supranational fiscal lever (common blockchain surpassing the 2021 OECD agreement)
- Global diffusion of the DCEN model (each member country as ambassador)

4.2 The concept of critical mass

The simulation showed that the 8-15 country threshold is decisive. Below 8, the coalition is symbolic. Above 15, it reaches the critical mass to impose standards at COP meetings and in trade agreements. This threshold was not determined theoretically - it emerged from the simulation itself.

4.3 Conditionality as an anti-free-rider mechanism

The identified risk: a country joins the coalition for the mutualised electricity grid without adopting the DCEN model. The solution - access conditioned on the fiscal floor rate and blockchain - is exactly the mechanism the EU already uses with its structural funds. This is not a theoretical innovation, it is an institutional transposition.

5. Integration of EducationLibre - Simulation 3

The integration of EducationLibre (github.com/nadnone/EducationLibre) into simulation 3 followed a specific method: read the source project, identify genuine synergies with the DCEN, and build the integration while respecting the original philosophy of EducationLibre.

5.1 Method of reading the source project

Three files were consulted directly from the GitHub repository:

File consulted	Key content extracted	Impact on the simulation
english/README.md	BitTorrent model, credit system, groups of 15 max, NVC, teacher validation	Basic structure of the integration into DCEN
french_main/README.md	Original French version, assumptions on 15 members, social contract	Philosophical nuances - patch of existing system, not replacement
english/FUNDAMENTALS.md	File being written - essentially empty at time of consultation	Indicates the project is still under construction - important for limits

5.2 Synergies identified between EducationLibre and DCEN

Two fundamental synergies were identified, beyond the obvious points of contact:

Philosophical isomorphism: BitTorrent = SSE

The BitTorrent model of EducationLibre (each member is simultaneously a client and server of knowledge) is structurally identical to the SSE philosophy (each member is simultaneously a producer and beneficiary). This is not a coincidence - it is the same logic of the commons applied to two different domains. This discovery structured the entirety of simulation 3.

The credit system as a solution to the governance-without-experts problem

Simulation 2 identified a gap: the SSE promised autonomous governance but lacked the mechanism to guarantee it. The EducationLibre credit system - which incentivises teaching what one knows - is precisely that mechanism. The self-learning SSE becomes possible.

6. Corrections made and their justifications

This chapter documents the important corrections made during the construction of the report. These corrections are not errors to hide - they are improvements arising from dialogue and must be documented for the model to remain intellectually honest.

6.1 Major correction: separation of credit systems

This is the most important correction in the report. In the initial version of simulation 3, a fundamental conceptual error had been made: EducationLibre credits were presented as a kind of European social currency, even convertible into DCEN economic value.

Before	After
<i>EducationLibre credits exchangeable between member countries create a cross-border social bond... The BitTorrent system gives rise to a European culture of epistemic solidarity.</i>	Credits NEVER circulate between members. They are transferred only to the managing cooperative via the mobile application. They are not convertible into real money.
Reason for correction:	Merging the two systems would have destroyed intrinsic motivation (Deci & Ryan). Teaching to earn money is not the same as teaching to share. The distinction is philosophically foundational.

Before	After
<i>Credits = exchange currency (in the DCEN/EducationLibre synergy table)</i>	Credits = internal reciprocity - independent of DCEN currency
Reason for correction:	The initial formulation implied fungibility between the two systems. The correction establishes their total impermeability.

6.2 Additions from dialogue: clarifications on concrete functioning

Three important clarifications were added following corrections raised by the project author:

Before	After
<i>P2P education as a digital-only system</i>	Three modes: virtual (online), in-person with connection, in-person offline with paper notebook synchronised at next connection. A single smartphone in the group is sufficient.
Reason for correction:	Offline mode is fundamental for including rural areas and populations without stable internet access. Omitting it would have reproduced exactly the divide the model seeks to correct.

Before	After
<i>Credits circulate between members according to a reciprocity logic</i>	Credits are ONLY transferred to the managing cooperative/association/foundation. Never between members. The mobile application is the interface for this transfer.
Reason for correction:	Allowing direct exchanges between members would create a credit market - speculation possible, inequalities reproducible. Transfer only to the managing structure eliminates this risk.

Before	After
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<i>The sovereign fund finances the EducationLibre infrastructure</i>	Explicit bootstrapping mechanism: external actors (SSE companies, municipalities, individuals) can BUY credits from the cooperative. This money funds certified institutional teachers to create the first network seeders.
Reason for correction:	Without a bootstrapping mechanism, the network stays blocked by the chicken-and-egg problem: no seeders without members, no members without seeders. Credit purchase by outsiders creates the first professional seeders.

7. Design principles retained

Through the iterations, several design principles imposed themselves as implicit guides. Documenting them here makes them explicit and criticisable.

Principle 1 - Systemic coherence before exhaustiveness

Each building block added was tested for coherence with the preceding ones before being accepted. A technically interesting but systemically incoherent element was rejected. The virtuous cycle - each element reinforces the others - was the primary validation criterion.

Principle 2 - Dialogue as a correction method

The most important errors (notably the merging of credit systems) were identified and corrected through dialogue with the project author. This confirms that this type of model cannot be built in isolation - it requires permanent confrontation with the holders of each component.

Principle 3 - Distinguishing steady state from transition

The DCEN model describes a desirable state. It does not say how to get there. This distinction was maintained throughout the report. The feasibility matrix (time horizons) is the only concession to the transition question - and it remains incomplete.

Principle 4 - Non-convertibility as a philosophical safeguard

The distinction between the EducationLibre credit system and DCEN currency illustrates a more general principle: some things should not have a price, not because they are worthless, but because monetising them would destroy them. Shared knowledge, care, and citizen governance belong to this category.

Principle 5 - Inclusion as a design constraint, not an objective

The inclusion of people in difficulty (unemployed, sick, those without internet) was not treated as an additional objective to achieve. It was treated as a design constraint: an element that only works for connected and available people was considered insufficient, regardless of its theoretical interest.

8. Methodological limits and potential biases

Every model has biases. Documenting them is a condition of its intellectual integrity.

Internal coherence bias

The model was built to be coherent with itself. This does not guarantee its coherence with economic and political reality. Economies operating on different principles (China, United States, Gulf countries) were not integrated as constraints - they might have invalidated certain building blocks.

Numerical simulation bias

The simulation figures (72% renewables, geopolitical weight score of 68, etc.) are relative indicators, not predictions. They show trends and relationships between variables - not absolute values. Presenting them without this caveat would be a serious methodological error.

Systemic optimism bias

The model assumes that the building blocks reinforce each other mutually and that actors cooperate in good faith. History shows that coalitions fragment, sovereign funds get captured, cooperatives bureaucratised. These risks are documented in the reports but not integrated as dynamic constraints in the model.

Limit: absence of modelling of political resistance

The model does not simulate the resistance of interest groups that lose in this scenario (fossil industry, tax havens, speculative funds). These actors have considerable resources and documented blocking strategies. Their absence from the model is an important limitation.

Limit: EducationLibre is still a project in progress

EducationLibre (github.com/nadnone/EducationLibre) is a utopian project under development, by its own description. Integrating it into a European simulation requires significant extrapolation from what currently exists. The report assumes this extrapolation consciously but it must be flagged as such.

9. Questions this work opens up

A good theoretical model does not close questions - it opens new ones. Here are the questions that building this report made more precise and which deserve further exploration.

Open question	Why it matters
Governance of the sovereign fund	Who decides on investments? What representation of the different stakeholders? How to prevent capture by partisan or financial interests? Democratic governance is asserted but not detailed.
Theory of transition	The model describes a steady state. How does one move from the current system to this model without a major crisis? What is the first concrete step? Which political coalition makes it possible in which country first?
Metrics for EducationLibre success	How does one measure that an EducationLibre network is functioning well? Number of members? Diversity of sub-subjects? Learner satisfaction? Autonomy of SSE cooperatives? These metrics have not been defined.
Interoperability between DCEN coalitions	If several regional coalitions adopt the DCEN model (Europe, Africa, Latin America), how do they interoperate? Are fiscal blockchains compatible? Are EducationLibre networks connected?
The role of AI in the EducationLibre network	AI tools could accelerate teacher validation, personalise learning pathways, or automatically translate sub-subjects. But they also risk recreating technological dependence and reducing the human dimension of the model. This tension has not been resolved.
Minimum viable scale of a DCEN cell	What is the minimum size of a functional DCEN cell? Can a municipality of 500 inhabitants operate the full model? Or is a minimum demographic threshold required? The geographic domino effect assumes cells are small enough to be reproducible.

Final note

This methodological document is itself a proposal, not a truth. It can and should be criticised, completed, contradicted. The value of a theoretical model is measured not by its perfection but by the quality of the discussions it provokes and the experiments it inspires.

Sources consulted: T. Piketty (2013, 2021) | E. Saez (2003-2024) | G. Zucman (2015) | EducationLibre - github.com/nadnone/EducationLibre (CC BY 4.0) | Deci & Ryan - Self-Determination Theory (1985) | M. Rosenberg - Nonviolent Communication (2003) | R. Feynman - Pedagogy (1963) | OECD - Global Tax Agreement (2021) | BitTorrent Protocol - B. Cohen (2001) | Wikimedia Foundation - funding model